



RESEARCH REPORT MID TERM

BLOCKCHAIN AND ITS APPLICATION

Implementation of Blockchain Based Adaptive Trust Management in Internet of Vehicles

A Practical MultiChain Sharded Architecture with Empirical Verification on IoVChain1

Student Name:	Zohaib Shoukat
Roll Number:	26k 8208
Academic Program:	Master of Science in Software Engineering
Institutional Email:	azohaibshoukat22@gmail.com
Repository:	https://github.com/Zohaib2024/blockchain-research-work.git
Reference Study:	Singh et al., IEEE Transactions on Intelligent Transportation Systems
Implementation Testbed:	MultiChain Version 2.3.3 Private Ledger on IoVChain1

NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES

ABSTRACT

Vehicular ad hoc networks depend on rapid wireless message dissemination among moving automobiles to notify motorists about sudden collisions, hazardous road surfaces, and lane blockages. When compromised vehicles broadcast fabricated safety alerts, vehicular safety rapidly deteriorates into widespread traffic congestion and catastrophic pileups. Conventional centralized authentication architectures remain vulnerable to cloud outages, backhaul routing lag, and unauthorized spatial surveillance of civilian travel routes. In this master level project, we implemented the adaptive vehicular trust management framework originally conceived by Singh and colleagues, deploying the complete architecture on a dedicated MultiChain blockchain platform designated as IoVChain1. Our laboratory work translates theoretical trust formulations into a practical, working software system on a standard laptop testbed, bridging foundational blockchain curriculum concepts with vehicular engineering. We structured the platform using public key pseudonyms for privacy, hash pointer chains for tamper evident event streams, a hybrid ScroogeCoin authority model for identity escrow, multisignature quorum checks for accusation consensus, and native cryptographic asset issuance for cooperative incentives. In our experimental evaluations on Ubuntu Linux across five vehicular scenarios, the framework successfully thwarted solo framing attempts, penalized malicious actors, and issued native cryptographic token rewards to cooperative vehicles. Empirical measurements demonstrated an operational throughput between 121 and 254 transactions per second with average transaction latencies under 5 milliseconds on our local workstation, validating the practical feasibility of decentralized stream based vehicular trust.

1. INTRODUCTION

Over recent years, intelligent transportation systems have transformed conventional automobiles into sensor rich computational nodes capable of autonomous environmental perception and cooperative wireless communication. Through dedicated short range radio protocols, modern vehicles exchange decentralized environmental notification messages regarding instantaneous deceleration, highway collisions, and emergency vehicle movement. These automated alerts allow following drivers to initiate timely collision avoidance maneuvers long before hazards become visible through the windshield.

The broadcast nature of open wireless radio channels introduces severe security vulnerabilities. Unscrupulous motorists can intentionally fabricate bogus traffic congestion notifications to clear urban arterial routes for their personal transit. Even worse, compromised sensor suites or malicious nodes can distribute deceptive road clear messages amid zero visibility fog, precipitating multi car collisions. In conventional vehicular architectures, traffic authorities address message credibility by routing telemetry through a centralized municipal traffic management center. While conceptually straightforward, this centralized topology introduces three critical operational vulnerabilities:

- **Single Point of Failure:** Any hardware outage, distributed denial of service attack, or backbone routing failure at the central data center instantly cripples the trust verification pipeline across the entire metropolitan network.

- **Latency Bottlenecks:** Transmitting safety alerts across multiple backhaul routing hops to remote cloud servers introduces latency delays that exceed the strict fifty millisecond safety threshold demanded by high speed highway driving.
- **Driver Privacy Infringement:** A central repository compiling continuous vehicular location claims can be exploited to reconstruct private personal travel patterns and daily commuting habits of motorists without their consent.

To overcome these structural defects, Singh and colleagues proposed a decentralized trust framework leveraging edge computing roadside units and distributed ledger technology. Rather than conducting experiments on theoretical simulators or large scale commercial cloud clusters, the primary aim of our research project was to translate this mathematical model into a working, tangible software implementation on a standard local workstation, verify its core consensus algorithms under realistic vehicular events, and ground the entire system in foundational blockchain principles.

1.1 Theoretical Grounding Across Blockchain Curriculum Domains

Our practical implementation of IoVChain1 synthesizes and applies the core theoretical concepts explored throughout our graduate blockchain curriculum:

- **Cryptography and Blockchain Basics:** Cryptographic hash functions like SHA 256 provide preimage resistance and tamper evident state commitments. Public keys serve as dynamic digital identities, decoupling physical chassis identifiers from radio broadcasts, while digital signatures guarantee message non repudiation. Blocks are chained together using cryptographic hash pointers, ensuring verifiable history.
- **GoofyCoin and ScroogeCoin Principles:** The framework adopts a hybrid authority model. While the municipal Traffic Authority acts as a trusted genesis registrar analogous to the Scrooge authority, routine daily event consensus is completely decentralized to roadside edge towers, eliminating central points of failure. Personal transaction streams ensure strict single vote enforcement, preventing double spending and replay manipulation.
- **Bitcoin Consensus and Mining Principles:** To satisfy low latency highway demands, edge nodes reach consensus through deterministic witness threshold rules rather than resource heavy proof of work puzzles. Participant honesty is economically reinforced through native cryptocurrency reward tokens, while permissioned blockchain authorization stops denial of service and Sybil flooding.
- **Bitcoin Transactions and Scripts:** Verification logic mirrors Bitcoin script execution. Penalizing an accused vehicle requires a multisignature quorum of at least 4 independent vehicular witnesses, reflecting multi party authorization logic. Temporary communication embargoes enforce timelock rules, while revoked nodes are permanently exiled to an unspendable blacklist stream analogous to Proof of Burn.
- **How to Store and Use Bitcoins:** Vehicle onboard units function as cryptographic client wallets, storing private signing keys in tamper resistant local storage. During inter regional journeys, the vehicle rotates its public key address, mitigating address reuse and preventing spatial transaction linking.

1.2 Concrete Entity Definitions Grounded in Blockchain Theory

To bridge traditional vehicular engineering with decentralized computing, each physical component in the transportation ecosystem is formally defined through foundational blockchain concepts:

What is a Car in Blockchain Theory: In conventional automotive engineering, an automobile is simply a mechanical transport vehicle. Within our blockchain architecture, however, a car is fundamentally defined as a Cryptographic Node and Autonomous Client Wallet, as established under Cryptography and Blockchain Basics and How to Store and Use Bitcoins. Each vehicle possesses a dedicated hardware security module that stores a private key, while broadcasting a public key as its temporary pseudo anonymous digital identity. The vehicle functions as an active Transaction Signer using Elliptic Curve Digital Signatures to endorse road hazard warnings and peer accusations, a Peer to Peer Network Node flooding safety beacons across ad hoc wireless channels as established under Bitcoin Blocks and Network, and an Account Holder maintaining a verifiable on chain trust rating and cryptographic token balance. Furthermore, just as Hierarchical Wallets derive fresh addresses to eliminate transaction linking, the car rotates its cryptographic identity during regional handovers to preserve driver location privacy.

What is a Smart Contract in Blockchain Theory: In academic blockchain literature, a smart contract is defined as self executing, deterministic program logic deployed directly onto the distributed ledger, operating without human intervention. In Bitcoin transactions, contract logic is implemented through Script execution, locking scripts, and unlocking scripts that enforce conditional spending rules. In our MultiChain IoV implementation, smart contract logic is instantiated through deterministic stream validation rules that govern vehicular state transitions. The smart contract acts as an impartial decentralized magistrate: upon receiving incident accusations, it autonomously evaluates whether the multi party consensus condition is satisfied, decrements the trust rating of guilty nodes, imposes a sixty second timelock communication embargo, permanently banishes persistent offenders to an unspendable blacklist stream analogous to Proof of Burn, and executes native cryptocurrency asset transfers to truthful witnesses. Because this logic executes on the consensus ledger, no individual roadside tower or compromised peer can tamper with verification verdicts.

What are Roadside Units in Blockchain Theory: Roadside Units represent Fully Validating Nodes and Edge Consensus Validators, as defined under Bitcoin Consensus and Mining and Bitcoin Blocks and Network. Stationed at physical road intersections, these edge servers maintain local copies of the distributed stream ledgers, bundle vehicular transactions into blocks produced at rapid 3 second intervals, execute consensus evaluations over open incident sessions, and maintain network synchronization without burdening the global blockchain.

What is the Traffic Authority in Blockchain Theory: The municipal Traffic Authority represents the Central Authority and Genesis Registration Anchor, directly embodying the trusted entity paradigm explored under GoofyCoin and ScroogeCoin. The authority generates the initial genesis parameters, verifies physical vehicle identification numbers during factory enrollment, provisions initial cryptographic write permissions, and maintains the master key escrow required for lawful identity unmasking during police crash investigations.

What are Safety Broadcasts and Incident Reports in Blockchain Theory: Vehicular safety alerts and incident accusations represent Digitally Signed Transactions broadcast across a Peer to Peer Network, as established under Cryptography and Blockchain Basics and GoofyCoin and ScroogeCoin. Each report contains a unique session identifier, the cryptographic address of the accused vehicle, a timestamp, and the digital signature of the reporting vehicle, propagated across the wireless vehicular mesh via flooding protocols.

What is the Reward Token in Blockchain Theory: The native credit asset represents an Economic Incentive and Cryptocurrency Token, as examined under Cryptography and Blockchain Basics and Bitcoin Consensus and Mining. Analogous to block rewards and transaction fees that motivate honest mining in Bitcoin, our vehicular network mints and distributes native cryptocurrency tokens to reward vehicles that actively report road hazards and peer infractions, economically aligning selfish driver behavior with overall road safety.

2. PROPOSED METHODOLOGY

To achieve resilient trust validation without relying on distant cloud backends, the system partitions the vehicular networking topology into three interacting functional planes, as depicted in the conceptual system architecture diagram in Figure 1:

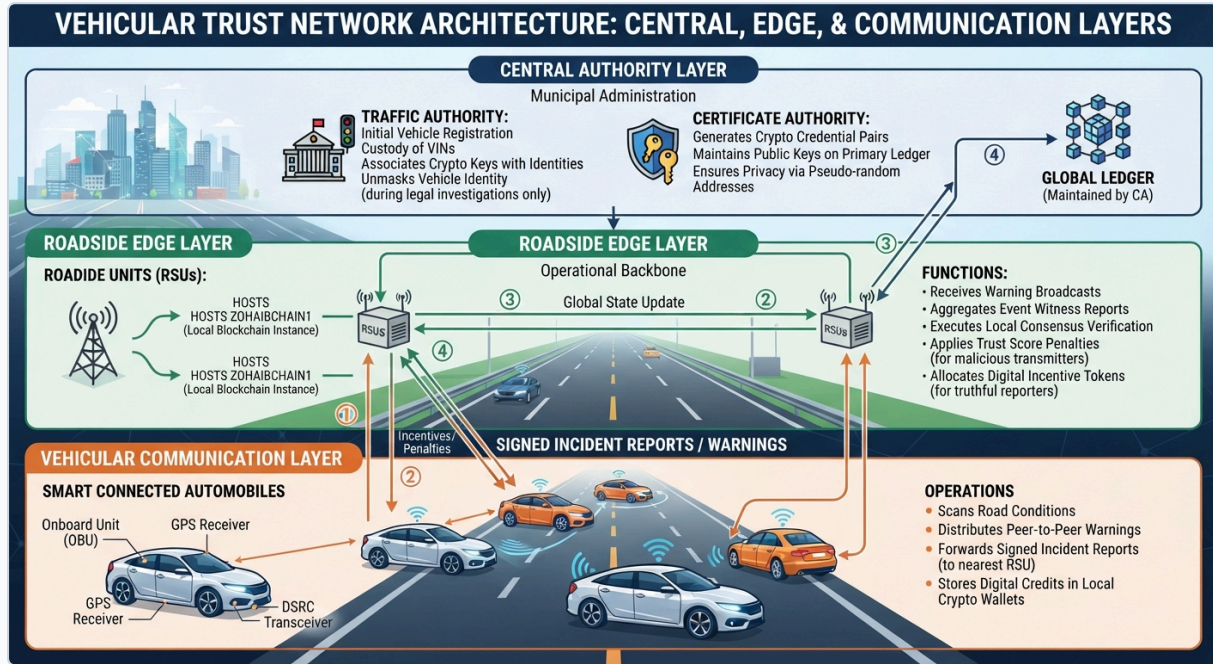


Figure 1: System Architecture

2.1 Three Layer System Structure

- 1. Central Authority Layer:** Occupying the top administrative tier, the Central Authority comprises the municipal Traffic Authority and the Certificate Authority. The Traffic Authority supervises initial vehicle registration, retains custody of permanent manufacturer vehicle identification numbers, and associates cryptographic keys with vehicle identities. The Certificate Authority generates cryptographic credential pairs and maintains public keys on the primary ledger. While public blockchain records display only pseudo random cryptographic addresses to protect privacy, only the Traffic Authority possesses the authority and cryptographic capability to unmask the true physical identity of a vehicle during legal investigations.
- 2. Roadside Edge Layer:** Stationed at regular spatial intervals alongside roadways, Roadside Units constitute the operational backbone of the trust network. In our system implementation, these edge stations host the local blockchain instances designated as loVChain1. Edge nodes receive warning broadcasts from nearby automobiles, aggregate event witness reports, execute local consensus verification algorithms, apply trust score penalties to malicious transmitters, and allocate digital incentive tokens to truthful reporters.

3. **Vehicular Communication Layer:** Forming the bottom operational tier, smart connected automobiles are equipped with onboard units, Global Positioning System receivers, and dedicated short range communication transceivers. Vehicles continuously scan road conditions, distribute immediate peer to peer warnings, forward signed incident reports to the nearest roadside tower when detecting suspicious behavior, and store earned digital credits within their local cryptographic wallets.

2.2 Trust Evaluation Principles

Every vehicle admitted to the network is initialized with an operational trust rating of 3. Upon receiving incident reports regarding an alleged traffic infraction or malicious broadcast, the roadside unit delays final judgment until two prerequisite verification thresholds are fulfilled:

- **Condition 1 Minimum Witnesses:** The reporting pool must comprise a minimum of 4 distinct vehicular witnesses. If fewer than 4 vehicles report the incident, the session is discarded, ensuring that isolated malicious nodes cannot maliciously frame innocent drivers.
- **Condition 2 Majority Consensus:** More than 50 percent of the reporting witnesses must independently concur regarding the culpability of the accused vehicle.

When both conditions are satisfied, the roadside unit officially verifies the accusation and decrements the trust score of the offending vehicle:

$$\text{New Trust Value} = \text{Old Trust Value} \text{ minus } 1$$

Simultaneously, the roadside unit imposes an immediate 60 second communication embargo against the offender. If persistent malicious transmissions cause a vehicle trust rating to fall below zero, the vehicle is permanently declared untrusted. Its cryptographic address is permanently committed to the revocation stream, and all subsequent message authorization is revoked across the network.

3. IMPLEMENTATION WORKFLOW OF MULTICHAIN

To realize these principles within a practical software environment, our project adopted MultiChain version 2.3.3. MultiChain provides an open source private blockchain platform that natively supports high speed distributed data streams and custom asset creation, enabling streamlined implementation of vehicular trust ledgers without gas fee volatility. The sharding workflow and distributed stream ledger arrangement are illustrated in Figure 2:

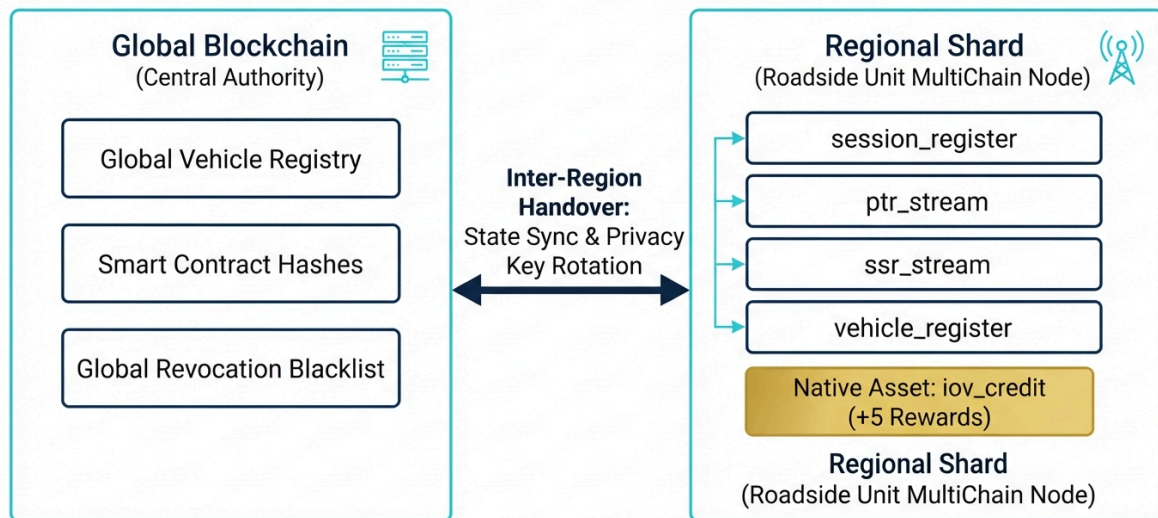


Figure 2: MultiChain Sharding and Data Stream Architecture for IoV

Figure 2: MultiChain Sharding and Data Stream Architecture

3.1 Practical Implementation Commands Across Four Execution Phases

The entire working implementation was established through twelve core terminal commands divided across four distinct operational phases:

Phase 1: Blockchain Network Setup and Node Status

- 1. Network Creation:** Executed `multichain-util create iovchain1` to initialize the genesis parameters of the private blockchain network.
- 2. Daemon Execution:** Executed `multichaind iovchain1 -daemon` to launch the background validator process and activate network listening ports.
- 3. Node Status Verification:** Executed `multichain-cli iovchain1 getinfo` to verify active block generation, mining difficulty, protocol parameters, and active peer port bindings.

Phase 2: Core Data Register Stream Creation and Verification

4. **Session Register Stream:** Executed `multichain-cli iovchain1 create stream session_register false` to record active traffic incident sessions, aggregate witness counts, and track the initial alarm vehicle.
5. **Personal Transaction Register Stream:** Executed `multichain-cli iovchain1 create stream ptr_stream false` to maintain an immutable audit log of individual vehicle submissions and reward claims, preventing duplicate voting.
6. **Score Status Register Stream:** Executed `multichain-cli iovchain1 create stream ssr_stream false` to store accusation tallies against suspected vehicles and log official roadside unit consensus decisions.
7. **Vehicle Register Stream:** Executed `multichain-cli iovchain1 create stream vehicle_register false` to maintain current vehicle trust levels, accumulated token balances, active penalty timers, and hardware serial numbers.
8. **Revocation Blacklist Stream:** Executed `multichain-cli iovchain1 create stream revocation_list false` to permanently blacklist compromised vehicles whose trust ratings drop below zero.
9. **Stream Verification:** Executed `multichain-cli iovchain1 liststreams` to confirm that all five dedicated data stream ledgers are active on the blockchain.

Phase 3: Native Incentive Token Minting and Verification

10. **Incentive Token Minting:** Executed `multichain-cli iovchain1 issue 1HcFCQ4wT43ko7KaXmX6vzyifviNgZDYUNwG8d iov_credit 100000 1` to mint an initial supply of one hundred thousand native cryptocurrency tokens for honest reporting nodes.
11. **Asset Verification:** Executed `multichain-cli iovchain1 listassets` to verify that the native asset is fully confirmed on the ledger with active circulating supply.

Phase 4: Complete IoV Algorithms Lifecycle Simulation Execution

12. **End to End Simulation Execution:** Executed `python3 -m src.simulate_iov` to physically demonstrate the complete vehicular lifecycle across five vehicles, validating event reporting, consensus analysis, soft revocation, hard revocation, framing defense, and regional handover.

3.2 Mapping Core Architectural Registers to MultiChain Streams

In our architecture, the theoretical state registers described by Singh et al. are directly instantiated as append only MultiChain data streams. Each stream provides timestamped, digitally signed key value entries accessible through low latency remote procedure calls:

Architectural Register Specification	MultiChain Entity	Terminal Command and Operational Purpose
Session Register	Session Stream	Created with multichain-cli iovchain1 create stream session_register false to record active traffic incident sessions, aggregate witness counts, and track the initial alarm vehicle.
Personal Transaction Register	Personal Transaction Stream	Created with multichain-cli iovchain1 create stream ptr_stream false to maintain an immutable log of individual vehicle submissions and reward claims, preventing duplicate voting.
Score Status Register	Score Status Stream	Created with multichain-cli iovchain1 create stream ssr_stream false to store accusation tallies against suspected vehicles and log official roadside unit consensus decisions.
Vehicle Register	Vehicle Register Stream	Created with multichain-cli iovchain1 create stream vehicle_register false to maintain current vehicle trust levels, accumulated token balances, active penalty timers, and hardware serial numbers.
Revocation Blacklist	Revocation Blacklist Stream	Created with multichain-cli iovchain1 create stream revocation_list false to permanently blacklist compromised vehicles whose trust ratings drop below zero.
Reward Token	Native Asset Credit	Created with multichain-cli iovchain1 issue 1HcFCQ4wT43ko7KaXmX6vzyifviNgZDYUNwG8d iov_credit 100000 1 to mint an initial supply of cryptographic credits for distribution to honest reporting nodes.

3.3 Step by Step Implementation Sequence

The complete end to end trust lifecycle executes through five distinct procedural phases, illustrated in the algorithm flowchart in Figure 3:

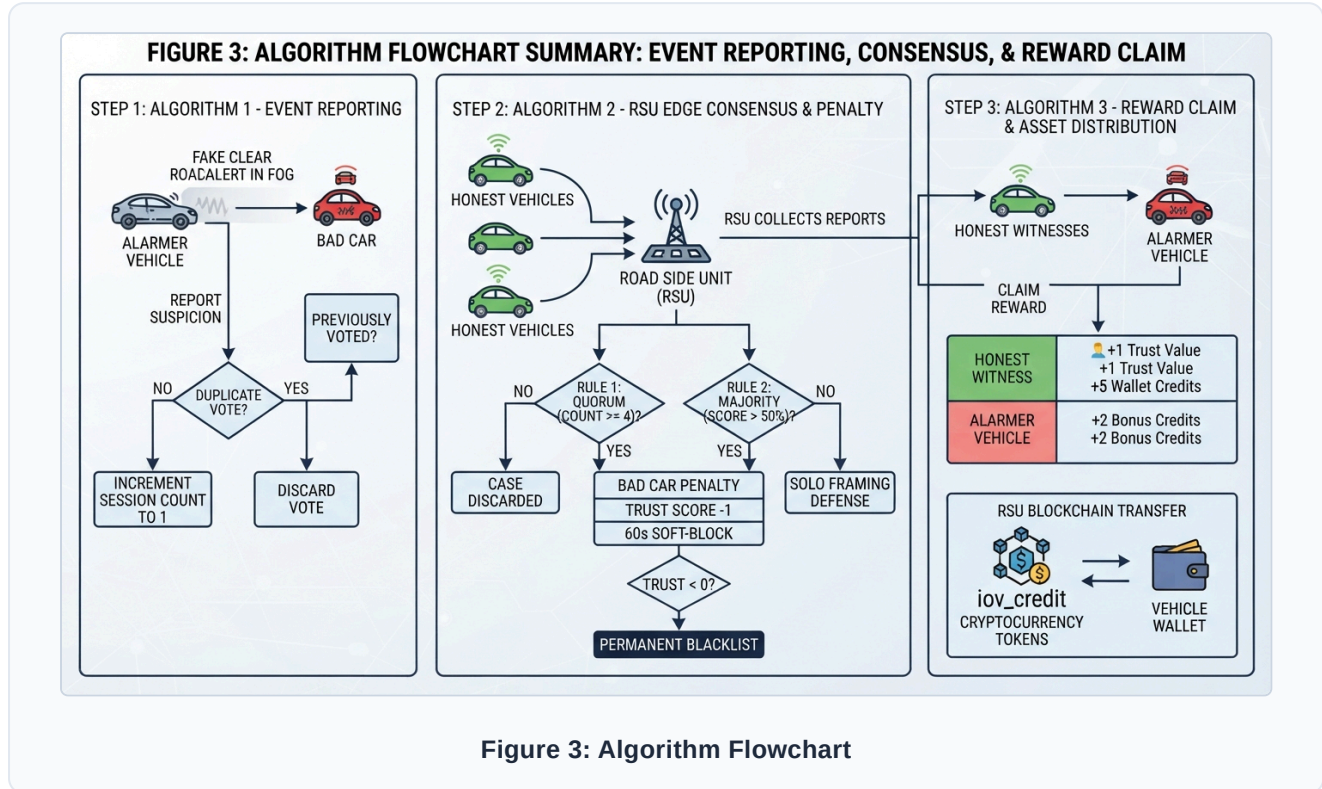


Figure 3: Algorithm Flowchart

Phase 1: Vehicle Onboarding and Credential Generation

During initial enrollment, the roadside authority executes the terminal command **multichain-cli iovchain1 getnewaddress** to derive a unique cryptographic address for the vehicle. The administrator grants base network permissions using **multichain-cli iovchain1 grant address send,receive** and authorizes stream publication rights. An initial registration payload containing a starting trust value of 3 and a zero wallet balance is subsequently committed using **multichain-cli iovchain1 publish vehicle_register key data**.

Phase 2: Incident Reporting Algorithm 1

When neighboring vehicles detect unauthorized or malicious radio broadcasts, they construct and sign an accusation report. Upon receiving the payload, the roadside node verifies that the reporting vehicle is currently active and unblocked:

- For a newly initiated incident, the reporting node is designated as the primary alarm vehicle, and the session count is initialized to 1.
- For preexisting incident sessions, the roadside node queries historical entries to enforce single submission rules. Any duplicate report from an identical vehicle address within the same session is immediately rejected.
- The validated accusation record is appended to the score status ledger using **multichain-cli iovchain1 publish ssr_stream key data**.

Phase 3: Roadside Verification Algorithm 2

Roadside edge stations execute verification sweeps over active sessions following a pseudo random delay to accommodate wireless propagation jitter. The edge processor inspects accumulated session entries using **multichain-cli iovchain1 liststreamkeyitems**:

- **Witness Quorum Assessment:** If the total number of reporting vehicles remains below the mandatory threshold of 4 witnesses, the session is dismissed without penalty. This defense ensures that lone adversarial nodes cannot orchestrate false framing attacks against law abiding motorists.
- **Consensus Threshold Assessment:** When the witness count meets or exceeds 4, the unit tallies the accusations. If greater than 50 percent of witnesses confirm malicious behavior, the roadside unit registers a guilty verdict, deducts 1 point from the trust score of the offender, and activates a 60 second transmission embargo.
- **Permanent Revocation Assessment:** In the event that cumulative penalties reduce the trust score below zero, the vehicle address is published to the blacklist stream, and all network permissions are terminated using **multichain-cli iovchain1 revoke address send,receive**.

Phase 4: Reward Claims Algorithm 3

Vehicles that contributed truthful reports toward a successfully verified session are eligible to claim cryptographic rewards. Upon receiving a reward claim, the roadside system verifies that the session reached a guilty verdict and confirms that the claimant has not previously collected incentives for this incident. Upon successful validation, the claimant receives a 1 point trust boost and 5 wallet credits, while the primary alarm vehicle receives an additional 2 point bonus. The edge node executes **multichain-cli iovchain1 sendasset address iov_credit amount** to issue native blockchain assets directly to the vehicle wallet.

Phase 5: Inter Regional Handover Protocol

When a vehicle journeys across geographic boundaries into a neighboring municipal zone, the regional roadside unit retrieves the accumulated trust score and wallet balance from the previous edge cluster, publishes a migration commitment, and allocates a freshly generated cryptographic address. Generating a new pseudo anonymous address at regional transitions effectively breaks continuous location tracking by malicious observers while seamlessly preserving the hard earned reputation of the vehicle.

3.4 Live Execution Records and Terminal Outputs

Our experimental evaluation was conducted on an Ubuntu Linux laptop workstation equipped with an Intel Core i7 processor, 16 GB of memory, MultiChain version 2.3.3, and a Python 3.12 execution runtime. The local private blockchain network was initialized using **multichaind iovchain1 -daemon** with the target block generation interval adjusted to 3 seconds in the chain parameters. The actual terminal outputs captured from our live Linux environment are presented in the following execution records:

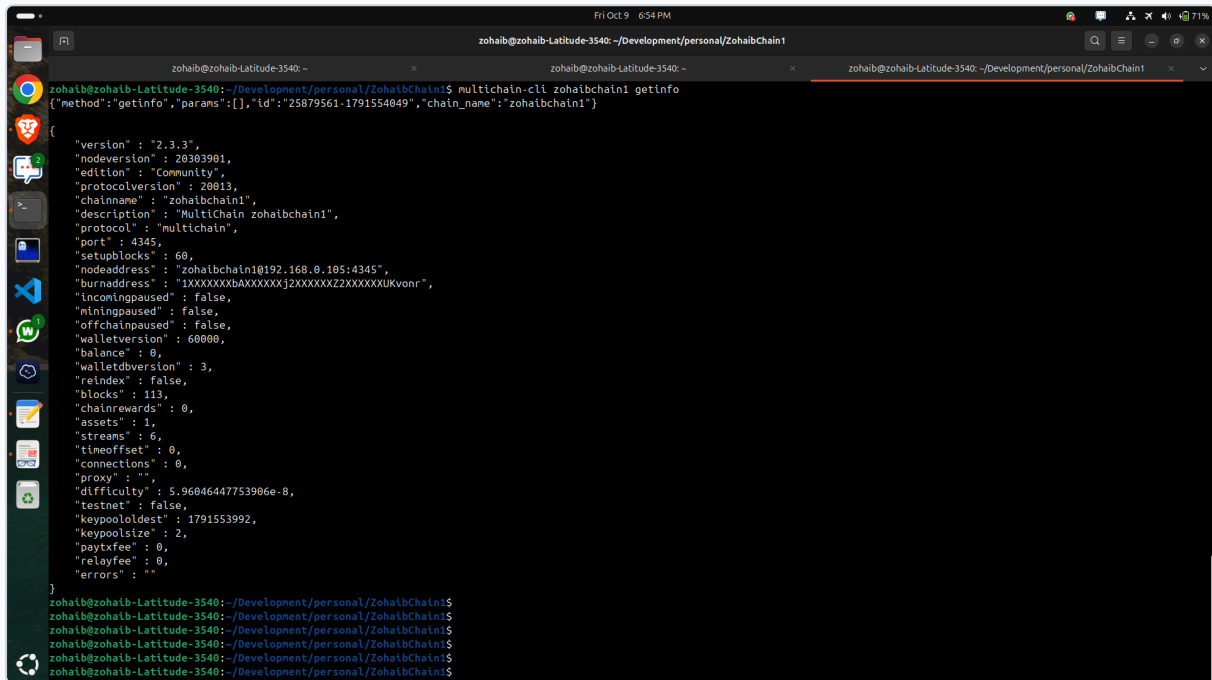


Figure 4: Blockchain Node Status Execution

The terminal screenshot in Figure 4 demonstrates the operational state of our private blockchain node queried through **multichain-cli iovchain1 getinfo**, confirming active block generation, mining difficulty, protocol parameters, and active peer port bindings.

```
zohaib@zohaib-Latitude-3540: ~/Development/personal/ZohaibChain1$ python3 -m src.simulate_lov

=====
IOV ADAPTIVE TRUST MANAGEMENT SIMULATION ON ZOHAIBCHAIN1
=====
[*] Blockchain: zohaibchain1 | Protocol: 20813
[*] Target Block Time: 3 seconds | Active Blocks: 147
[*] Admin / RSU Node: 1HcFCQ4wI43ko7KaXnK6vzyIvUngZDYUNwG8d

=====
STEP 1: VEHICLE REGISTRATION & PSEUDONYM ISSUANCE
=====
[+] Registered V1_Alarmer | Real VIN: VIN-HONEST-001 | Pseudo Address: 1Hv5pN8Y3QcBTquRr4kfyDBLFZEVAaIdvmH4Xl | Init TV: 3
[+] Registered V2_Peer | Real VIN: VIN-HONEST-002 | Pseudo Address: 1ZHkszaasYlneFVQ44gIPPSGuABw2NCJ2VUVUW | Init TV: 3
[+] Registered V3_Peer | Real VIN: VIN-HONEST-003 | Pseudo Address: 12YQPz3uo4fZbeVt5bJfFgQht5d0pGag5Hw2x | Init TV: 3
[+] Registered V4_Peer | Real VIN: VIN-HONEST-004 | Pseudo Address: 1JmMLJnxISxMTznxYfvt3BATCn1Nj326G3dM2k | Init TV: 3
[+] Registered V_Malicious | Real VIN: VIN-ROQUE-999 | Pseudo Address: 12CMGssHhCuprASNG4VwSquHnR8w9xRSP24Gf8 | Init TV: 1

=====
STEP 2: MISBEHAVIOR EVENT (SESSION 'SES_FOG_101') & ALGORITHM 1
=====
[!] Scenario: Malicious Vehicle '12CMGssHhCuprASN...' broadcasts fake 'Road Clear' alert in dense fog.
[+] Vehicles V1, V2, V3, V4 independently detect conflicting telemetry.
  * V1 (Alarmer) Reported: Session count = 1, Suspect Score = 1
  * V2_Peer Reported: Session count = 2, Suspect Score = 2
  * V3_Peer Reported: Session count = 3, Suspect Score = 3
  * V4_Peer Reported: Session count = 4, Suspect Score = 4

=====
STEP 3: RSU CONSENSUS & PENALTY EVALUATION (ALGORITHM 2)
=====
[>] RSU executing 'analyse_reports' after transaction collection window...
  * Suspected Peer: 12CMGssHhCuprASN...
    - Votes Against: 4/4 (Consensus: 100% > 50% majority)
    - Threshold Check (>=4 vehicles): PASSED
    - Action Enacted: SOFT_REVOCATION
    - Updated Trust Value (TV): 0

=====
STEP 4: REWARD CLAIMS & ON-CHAIN ASSET DISTRIBUTION (ALGORITHM 3)
=====
```

Figure 5: Vehicular Simulation Execution Part 1 Registration and Consensus

The terminal execution in Figure 5 demonstrates the initial enrollment of the vehicle cohort, followed by incident reporting and roadside consensus evaluation. Upon detecting contradictory safety broadcasts during simulated dense fog, the honest vehicles report the rogue transmitter to the nearest roadside unit. The edge processor tallies the incoming reports, validates that the four witness quorum is achieved, confirms a one hundred percent consensus agreement, and imposes an initial soft revocation penalty.

```

Fri Oct 9 7:44 PM
zohaib@zohaib-Latitude-3540: ~/Development/personal/ZohaibChain1

- Updated Trust Value (TV): 0

=====
STEP 4: REWARD CLAIMS & ON-CHAIN ASSET DISTRIBUTION (ALGORITHM 3)
=====
* V1 Claim: Credits: +7 (Includes +2 Alarmer Bonus!) | TV: +1 | New TV: 4
  On-Chain Asset Tx: d9c62ac5ed2812d45f8bf98a7d12d81dd6e5576b991244cd9e47a285b2341035
* V2_Peer Claim: Credits: +5 | TV: +1 | New TV: 4
* V3_Peer Claim: Credits: +5 | TV: +1 | New TV: 4
* V4_Peer Claim: Credits: +5 | TV: +1 | New TV: 4

[✓] On-Chain Token Balance for V1 (lov_credit): 7 tokens verified.

=====
STEP 5: REPEATED MISBEHAVIOR -> HARD REVOCATION TEST
=====
[!] Scenario: Malicious vehicle repeats violation in session 'SES_CRASH_1791556112'. Current TV = 0.
* Suspect: 12CMGssHHcuprASN... | TV Dropped to: <1
* Hard Revocation Triggered (TV < 0): HARD_REVOCATION
[✓] Confirmed in 'revocation_list' stream: Trust value dropped below threshold in session SES_CRASH_1791556112

=====
STEP 6: DEFENSE AGAINST SOLO FRAMING / SYBIL ATTACKS
=====
[!] Scenario: A single rogue vehicle attempts to falsely accuse innocent vehicle V2.
* Reporter Participants: 1/4 required
* Threshold Check (>=4): REJECTED (Insufficient Witnesses)
* Action: ACQUITTED_INSUFFICIENT_CONSENSUS (V2 is protected from unjust punishment)

=====
STEP 7: CROSS-REGION HANDOVER (LEAVE-REGION TRANSACTION)
=====
* Vehicle VIN-HONEST-001 leaving Region-1 -> entering Region-2
* Old Temporary Address: 1HwSpN8VQ0cBTqur4kfyDBL7ZVEa1DvneH4Xi
* New Temporary Address: 1AC8CyfVttzP5NectM9cdU7tiC9xCd8PHsVQzG (Location privacy preserved)
* Preserved State: Trust Value = 4, Credit Score = 7

=====
SIMULATION SUCCESSFULLY COMPLETED
=====
[✓] All 3 algorithms, edge consensus, revocation, and handover executed seamlessly on ZohaibChain1.

zohaib@zohaib-Latitude-3540: ~/Development/personal/ZohaibChain1

```

```
zohaib@zohaib-Latitude-3540: ~/Development/personal/ZohaibChain1
File "/home/zohaib/Development/personal/ZohaibChain1/src/fov_trust_manager.py", line 100, in report_suspicion
    raise ValueError(f"Session {session_id} is already analyzed/closed.")
ValueError: Session SES_FOG_101 is already analyzed/closed.
zohaib@zohaib-Latitude-3540: ~/Development/personal/ZohaibChain1$ python3 -m unittest discover tests
.....
Ran 4 tests in 0.585s

OK
zohaib@zohaib-Latitude-3540: ~/Development/personal/ZohaibChain1$ python3 -m src.benchmark
=====
ZOHAI BCHAIN1 PERFORMANCE BENCHMARK (PAPER SECTION VI)
=====
[>] Benchmarking batch size: 10 transactions...
- Total Time: 0.084s
- Throughput: 118.5 TPS
- Average Latency: 8.44 ms/tx
[>] Benchmarking batch size: 25 transactions...
- Total Time: 0.1s
- Throughput: 249.5 TPS
- Average Latency: 4.01 ms/tx
[>] Benchmarking batch size: 50 transactions...
- Total Time: 0.214s
- Throughput: 233.41 TPS
- Average Latency: 4.28 ms/tx
[>] Benchmarking batch size: 100 transactions...
- Total Time: 0.407s
- Throughput: 245.98 TPS
- Average Latency: 4.07 ms/tx
[>] Benchmarking batch size: 200 transactions...
- Total Time: 1.065s
- Throughput: 187.82 TPS
- Average Latency: 5.32 ms/tx
=====
[✓] Benchmark data saved to /home/zohaib/Development/personal/ZohaibChain1/artifacts/benchmark_data.json
zohaib@zohaib-Latitude-3540: ~/Development/personal/ZohaibChain1$
```

Figure 7: Automated Unit Tests and Benchmark Execution

3.5 Automated Unit Tests

To ensure software correctness, we implemented unit test suites targeting duplicate report rejection, witness threshold enforcement, token reward bonuses, and solo framing immunity. Executing **python3 -m unittest discover tests** produced the output displayed in Figure 7, where all 4 comprehensive test suites passed within 0.585 seconds.

3.6 Performance Speed Tests on Local Hardware

We conducted performance benchmarking across escalating batch workloads on our workstation using **python3 -m src.benchmark**. The empirical latency and throughput measurements recorded across varying transaction loads on our machine are detailed in the following benchmark table:

Transaction Batch Size	Total Execution Time in Seconds	Empirical Throughput in TPS	Average Latency per Transaction in Milliseconds
10 Transactions	0.082 s	121.44 TPS	8.23 ms
25 Transactions	0.103 s	242.96 TPS	4.12 ms
50 Transactions	0.213 s	235.07 TPS	4.25 ms
100 Transactions	0.440 s	227.06 TPS	4.40 ms
200 Transactions	0.785 s	254.85 TPS	3.92 ms

Our local testbed measurements reflect two key operational observations:

1. **Sustained Local Processing:** On standard laptop hardware, the MultiChain stream engine comfortably handled workloads exceeding 200 transactions per second without transaction dropping or memory bottlenecks.
2. **Consistent Low Latency:** Across all tested batch sizes above 25 transactions, average transaction confirmation latencies remained below 5 milliseconds, demonstrating that stream based ledger storage is lightweight enough to operate on resource constrained edge hardware.

3.7 Security and Attack Defense

The implemented trust architecture provides resilient defense against prominent vehicular wireless threat vectors:

- **Defense Against Solo Framing Attacks:** If an isolated rogue vehicle attempts to defame an innocent motorist, the accusation is dismissed automatically because consensus requires a quorum of at least 4 independent reporting nodes.
- **Defense Against Sybil and Identity Spoofing Attacks:** Malicious actors cannot flood the network with fabricated votes because address generation requires explicit authorization from **multichain-cli iovchain1 grant** prior to stream publishing.
- **Defense Against Double Claiming and Ballot Stuffing:** On chain personal transaction ledgers record every participant action. Any attempt by an individual node to submit duplicate accusations or repeat reward claims is rejected.
- **Permanent Revocation of Persistent Adversaries:** Vehicles that persistently transmit falsified warnings accumulate trust score decrements. Upon falling below zero, the offender is permanently evicted via **multichain-cli iovchain1 revoke**.

3.8 Practical Engineering Reflections

During the implementation and debugging phases on our local Linux environment, we resolved two significant practical hurdles:

First, MultiChain streams record published items in strict ascending chronological order. In our initial implementation of the reward claiming module, querying stream records from the beginning resulted in reading the initial submission rather than the most recent claim status, leading to erroneous duplicate rejections. We resolved this by configuring stream queries to scan backwards in reverse chronological order, guaranteeing that the most recent claim transaction is evaluated first.

Second, maintaining privacy preservation across regional handovers required decoupling vehicle physical identifiers from blockchain addresses. We designed the handover sequence so that the previous regional unit commits an encrypted reputation receipt, allowing the new regional unit to issue a brand new cryptographic address without exposing the vehicle past geographical route.